

Shubham Kumar Gupta

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Work Experience

- **AI Coding Tutor (Software Engineering Expert) at xAI (Previously Twitter)** (Dec'24 - Present)
 - Preparing novel software engineering **dataset** to train the **Grok's** new model.
 - Responsible for writing **instructions, test cases and implementation** of problem related to **various domains in software engineering** with 100% accuracy and good enough for the **LLM** to train on.
- **Senior Software Engineer at InstaService Inc** (Jul'23 - Nov'24)
 - Developed a **Retrieval-Augmented Generation (RAG)** system for a live chatbot in **Python**, ensuring compatibility with various **Large Language Models (LLMs)**, improving response accuracy by **30%**.
 - **Architected** and deployed the **complete infrastructure** across **3** different environments along with **CI/CD pipelines**.
 - Deployed and managed **five backend services, three UI services, one messaging queue, and two database services**.
 - Implemented comprehensive **observability and monitoring systems** to track downtime and performance.
 - **Fine tuned OpenAI models** as per needs. Integrated those models into the **RAG System** for better accuracy.
 - Made one of the **services secure** by implementing suitable **authentication** for all of its endpoints. It was open initially.
 - Created a service provider **referral system** in **Node.js**. Developed dashboard APIs for business owners.
- **Senior Engineer at Cognius.ai** (Jun'21 - Jul'23)
 - Developed **Python** and **NodeJS microservices** from inception following thorough requirements analysis and documentation. One of these services serves as the **core microservice** for the entire product, responsible for computing embeddings for provided inputs.
 - Enhanced robustness and performance of a **Python** microservice, achieving a **10x** performance improvement. Integrated **GPT-3, GPT-4, and ChatGPT** models into the same service, optimizing them for CPU execution. Overall product optimization facilitated its operation on a macOS system with 8 cores.
 - Architected platform infrastructure with **Kubernetes, Messaging Queue, Virtual Machines (VMs), and Load Balancers (LBs)** on **Google Cloud Platform (GCP)** using **Terraform** and **Ansible**.
 - Oversaw maintenance and scalability of the entire platform architecture, ensuring it could handle **1000 requests per second (req/sec)**.
 - Contributed to the generation, creation, and fine-tuning of **training data** and bots for **Hindi** language applications. The established bot setup serves as a standard for all Hindi language demonstration purposes.
 - Led on-premises and cloud deployments for various clients across multiple channels including **Google Cloud Platform (GCP), Microsoft Azure, and Amazon Web Services (AWS)**.
 - Tailored GPT-3 models to align with organizational requirements, resulting in a **20%** increase in bot response effectiveness.

Academic Qualifications

Degree/Certificate	Institute	Year	CPI/%
M.Tech.	Indian Institute of Technology Kanpur	2019-2021	7.18/10
B.Tech. (Computer Science & Engg.)	Madan Mohan Malaviya University of Technology	2014-2018	7.45/10
Intermediate (XII)	Divine Public School (CBSE)	2013	84.2 %
High School (X)	SDDT Intermediate College (CBSE)	2011	9.4/10

Technical Skills

- **Languages:** C . C++ . Python . JavaScript . HTML5 . CSS3 and Java.
- **Tools/Frameworks/Runtime Environments:** Flask . Postgres . MongoDB . Elasticsearch . MLOps . CI/CD . Prompt Engineering . GPT-3 . GPT-J . Git . Jenkins . Docker Products . Kafka . RabbitMQ . Ansible . Terraform . API Development . Backend . JMeter . Load Testing . DevOps . Google Cloud Platform (GCP) . Kubernetes . Node.js . Deep Learning . Natural Language Processing (NLP) . LangChain . L^AT_EX.

Scholastic Achievements and Extra-Curricular Activities

- **5 star** coder on Hackerrank in **problem solving, C++ and Python** tracks.
- Scored **97.53 percentile** in **GATE CSE'19**
- Secured **AIR 553** in **Malaviya Entrance Test(MET)'14**
- **Top 2 percent** in "Introduction to modern application development" course organized by **NPTEL and IIT-M**.
- **Certificate of Excellence** from VC of university for designing and developing E-Cell website.
- Won goodies and T-Shirt in Hacktober Fest organized by **digiocean** for contribution to **open source**.